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Optics & Laser Technology

Date: January 2026

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Interplay of microstructure and mechanical properties in vacuum laser beam welded 50-mm-Thick titanium alloy Plates: Insights from molten pool flow simulation

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10.1016/j.optlastec.2025.114187 ↗

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Highlights

- Combined experiment and simulation on 50-mm-thick Ti alloy VLBW.
- A gradient of refined grains is formed along the thickness in the fusion zone.
- Molten pool flow is governed by thermal gradients and recoil pressure.
- Fusion zone is strengthened by martensite interfaces, enhancing strength and hardness.

Abstract

Vacuum laser beam welding (VLBW) offers significant potential for fabricating thick-section aerospace load-bearing structures, demonstrating capabilities comparable to electron beam welding. However, the origins of through-thickness microstructural heterogeneity and its impact on mechanical properties remain poorly understood. Here, we combine experiments and simulations to investigate how vacuum-mediated fluid flow and heat transfer govern energy distribution, microstructure, and mechanical gradients in 50mm-thick TC4 titanium alloy VLBW joints. Results reveal that the weld seam exhibits distinct deep-penetration characteristics. Although the grain morphology remains largely uniform from top to bottom, the grain size displays a clear gradient, decreasing from $2.59\mu\text{m}$ at the top to $2.12\mu\text{m}$ at the bottom. Tensile fractures consistently occurred in the base metal (BM), confirming joint strength superiority. Simulations attribute these effects to deep keyhole formation with minimal spatter and steep thermal gradients. Recoil pressure and Marangoni convection drive spatially heterogeneous melt flow, causing non-uniform heat transfer and thermal distribution. This produces distinct microstructures across weld regions (top-TR, middle-MR, bottom-BR). Our work provides mechanistic insights and experimental validation for optimizing thick-plate VLBW processes and enhancing joint performance.

Keywords

Vacuum laser beam welding; Thick plate welded joint; TC4 titanium alloy;
Microstructural gradient; Molten pool flow behavior

5. Conclusion

CRedit authorship contribution statement

Declaration of competing interest

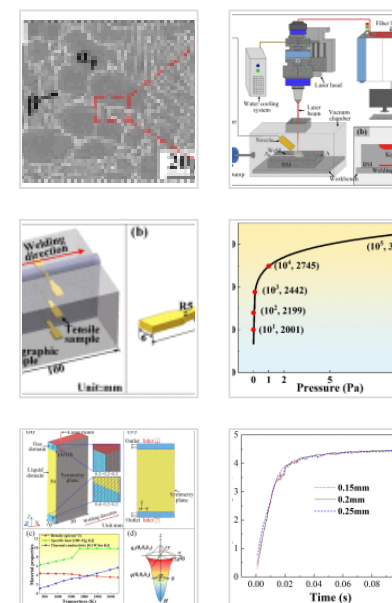
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1. Introduction

Titanium alloys are widely recognized for their superior mechanical properties and high strength-to-weight ratio. These distinctive attributes have led to their extensive use in the aerospace and military sectors, establishing titanium alloys as an essential material in the construction of aircraft load-bearing beam structures [1,2]. With the expanding applications of titanium alloys, the demand for thick titanium alloy plates in structural and load-bearing components has significantly increased [3,4]. However, thick plate welding often encounters defects such as porosity, cracking, and deformation, coupled with a large heat-affected zone (HAZ), making it challenging to balance welding efficiency and post-weld performance [5,6].

Electron beam welding (EBW) has partially addressed these issues, offering advantages like concentrated energy, deep penetration, and a narrow HAZ, which enable highly efficient welding [[7], [8], [9]]. Nevertheless, EBW is not without its limitations, including complex equipment, stringent vacuum requirements, and susceptibility to surface oxidation in the weld zone. To further enhance the welding quality of thick titanium alloy plates, vacuum laser beam welding (VLBW) has emerged as a promising alternative. By integrating the high energy density of laser welding with the benefits of a vacuum environment, VLBW not only mitigates the equipment complexity and oxidation issues associated with EBW but also enables superior welded quality control and enhanced post-weld properties. This advancement provides a more efficient and reliable solution for welding thick titanium alloy plates, marking a significant step forward in this critical field [[10], [11], [12]].

Under the vacuum environment, the plasma plume generated by high-energy-density laser can be effectively reduced, thereby improving the utilization rate of laser energy [13]. Simultaneously, convective heat transfer is eliminated in a vacuum, with heat transfer relying solely on conduction and radiation, which occurs at a relatively slower rate. This results in a significant difference in cooling rates compared to atmospheric conditions, thereby influencing the welding structure and flow state [[14], [15], [16]]. In recent years, there have been some studies on VLBW of TC4 titanium alloy and other materials. Li et al. [17] found that VLBW of TC4 titanium alloy thick plates enhanced joint

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strength through α'/α' strengthening at martensite interfaces, achieving an average tensile strength of 1010MPa. Krichel et al. [18] investigated VLBW of S355JR steel, finding that higher laser power, slower welding speed, and a negative defocus increased weld penetration. Wang et al. [19] found that heat treatment of TC4 titanium alloy welded by VLBW transformed the microstructure into acicular α and $(\alpha+\beta)$ phases, reducing tensile strength from 1009MPa to 586MPa. Gumenyuk et al. [20] achieved 28mm vacuum laser single-pass welding of S355 steel with an electromagnetic molten pool. They observed that higher speed caused hardening and cracking, while lower speed resulted in larger grains, both reducing joint performance. Li et al. [21] realized deep-penetration welding of 100mm-grade thick titanium alloy is achieved using vacuum laser beam welding, with the joint strength reaching 987MPa. Jiang et al.[22] investigated vacuum welding of 20 mm thick Al alloy with a beam oscillation technique and found that beam oscillation significantly reduced porosity, greatly enhancing the mechanical performance of the thick alloy joints. However, due to challenges in experimental observation and the complexity of melt pool dynamics, the internal flow behavior within the weld pool across different regions in the thickness direction has not yet been fully elucidated. Further in-depth research is essential to advance our understanding in this area.

Numerical simulation plays a pivotal role in advancing the understanding of welding processes for thick titanium alloy plates. The bulk of existing numerical studies has concentrated on laser welding of titanium alloys, with Zhang et al. [23] developed a precise micron-scale computational model for TC4 titanium alloy laser welding, identifying key process parameters influencing keyhole morphology. Chen et al. [24] established a numerical model for laser welding of TC4 titanium alloy to study protective gases, finding that increasing gas flow rate effectively suppresses plasma plume and improves laser utilization. Currently, research on the numerical simulation of VLBW remains relatively limited, primarily due to the inherent complexity of experimental conditions and the significant challenges associated with model development. To date, investigations by researchers such as Gong et al. [25] have developed a model for keyhole dynamics and melt flow in VLBW of Ta-10W alloy. They found that optimizing boiling

point and evaporation pressure in a vacuum reduced keyhole collapse frequency and amplitude, enhancing stability and minimizing closure. Han et al. [26] conducted numerical simulations to investigate the deep penetration characteristics of laser deep penetration welding under vacuum conditions. The enhanced model incorporates the effects of ambient pressure and the laser-induced plume, and the simulation results show excellent agreement with experimental data. Chen et al. [27] investigated vacuum oscillating laser welding of zirconium alloys, using beam oscillation and numerical simulations to elucidate the molten pool flow behavior and the periodic evolution of the keyhole during the welding process. This necessitates further VLBW numerical simulations to reveal the evolution of microstructures and flow behaviors in different regions of the weld seam, thereby providing a theoretical foundation and practical guidance for optimizing welding processes and enhancing weld quality.

To gain an in-depth understanding of the microstructural and mechanical property gradients in TC4 titanium alloy thick plates during the VLBW process, a systematic investigation was carried out on the microstructure and mechanical behavior. Meanwhile, a computational fluid dynamics (CFD) model tailored for thick-plate TC4 VLBW was developed to analyze the temperature and velocity fields in the top (TR), middle (MR), and bottom (BR) regions of the molten pool. This was aimed at elucidating the influence of flow characteristics on the microstructural evolution and mechanical performance of the welded joints.

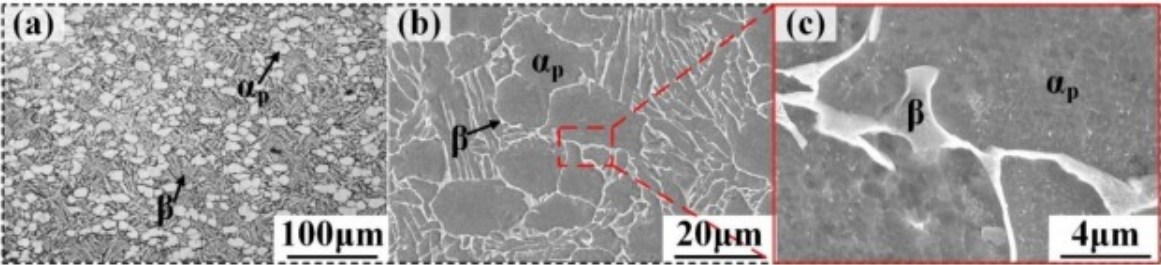
2. Materials and experiments

In this study, TC4 titanium alloy (Ti-6Al-4V) was employed as the primary material, with its detailed chemical composition presented in Table 1. The microstructure of the TC4 titanium alloy base metal (BM) exhibited a typical $\alpha+\beta$ dual-phase structure, as illustrated in Fig. 1. Two distinct heterogeneous structures were identified within the material: β phase, characterized by a body-centered cubic crystal structure, and the primary α phase, which adopts a hexagonal close-packed structure [28]. Observations showed that the primary α phase exhibited an equiaxed grain morphology, with its grain size significantly

larger than that of the adjacent β phase. In contrast, the β phase exhibited an elongated columnar morphology, distributed along the grain boundaries of the α phase [29].

Table 1. The chemical composition of Ti-6Al-4V titanium alloy.

Element	Al	V	Fe	C	N	H	O	Ti
Weight%	5.5~6.8	3.5~4.5	≤0.30	≤0.10	≤0.05	≤0.015	≤0.20	Bal.



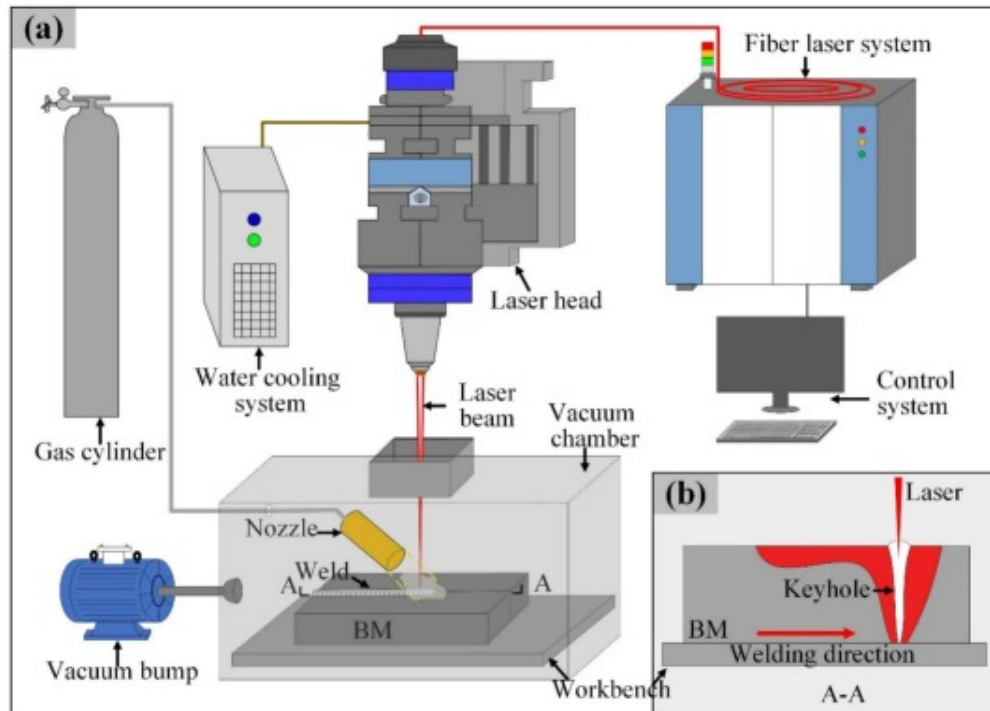
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Fig. 1. Microstructure image of the BM: (a) The OM image; (b) The SEM image; (c) Enlarged view of the local area in Fig. 1(b).

To further evaluate the welding performance of the material, two welding specimens with dimensions of 100 mm×40 mm×50 mm were prepared for single-pass welding experiments using a butt joint configuration. Prior to welding, the specimen surfaces underwent standardized pretreatment to ensure optimal results. This process involved mechanical polishing with sandpaper followed by chemical cleaning using acetone to effectively remove surface impurities and oxide layers. The welding was performed in a controlled vacuum environment, where pretreated specimens were placed inside a vacuum chamber. The chamber pressure was reduced to approximately 1000 Pa using a vacuum pump before initiating the laser welding process. A schematic representation of

the VLBW system and the process was provided in Fig. 2. The optimized welding parameters, determined through a comprehensive review of the literature[17,18] on VLBW of TC4 thick plates and complemented by preliminary experimental investigations, are adopted in both atmospheric and vacuum conditions and are detailed in Table 2.



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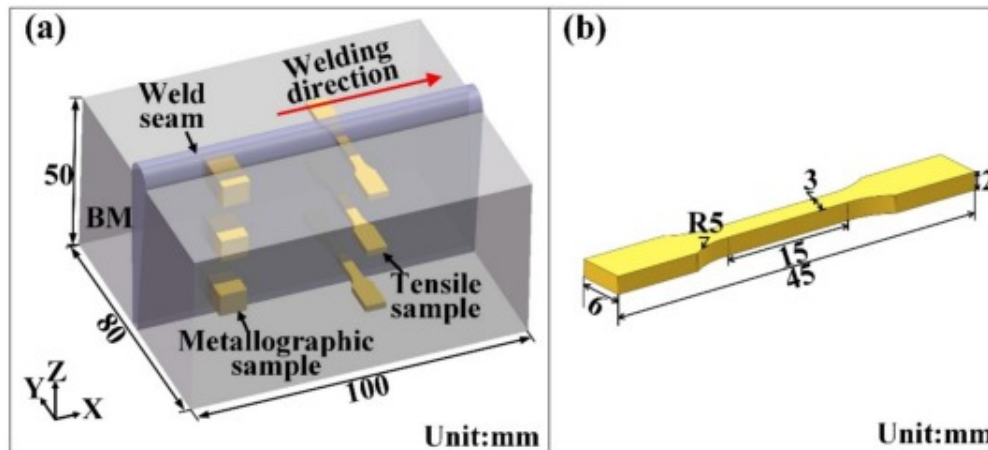
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Fig. 2. The schematic diagram of experimental setup for VLBW process: (a) Experimental setup; (b) VLBW process.

Table 2. The welding process parameters of Ti-6Al-4V titanium alloy.

Laser power (kW)	Welding speed (m/min)	Defocusing distance (mm)	Protective gas flow (L/min)	Vacuum degree (Pa)
16	0.3	−1	20	1000

To characterize the microstructure and mechanical properties of the material, specimens for microstructural and tensile analysis were extracted from distinct regions of the weld, as illustrated in Fig. 3(a). Specifically, both metallographic and tensile samples were collected from the TR, MR, and BR of the weld to evaluate the variations in microstructure and mechanical properties across these regions. The specimens were precisely cut using electrical discharge machining. Fig. 3(b) provides detailed dimensions of the dog-bone-shaped tensile specimens, which were designed in accordance with the GB/T 2651–2008 standard, with the tensile loading direction perpendicular to the weld seam. Following sectioning, the metallographic specimens were sequentially ground using 200#, 400#, 800#, 1500#, 2000#, and 3000#-grit molybdenum sandpaper, polished to a mirror-like finish, and subsequently etched with Kroll's reagent (5 mL HNO₃, 2 mL HF, 100 mL H₂O) to reveal the microstructural features. Optical microscopy (OM, MR-5000) was employed to examine the microstructure of the TR, MR and BR of the fusion zone (FZ). Scanning electron microscopy (SEM, HITACHI S-5000) was utilized to investigate the microstructure and fracture morphology, while electron backscatter diffraction (EBSD, JSM-7900F) was conducted to analyze grain size, texture orientation, and phase composition. The EBSD data were processed using the AztecCrystal software. Tensile properties and hardness were measured using a universal testing machine (UTM-5105) and a Vickers hardness tester, respectively.



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Fig. 3. Schematic diagram of sample position and size: (a) Sample position; (b) Tensile specimen sizes.

3. Numerical simulation modeling

3.1. Assumptions in the simulation

During the simulation of VLBW, material properties were temperature-dependent, and the process involved intricate physical phenomena such as phase transformation, evaporation, and fluid dynamics. To ensure computational accuracy while maintaining model simplicity, certain assumptions were introduced. The specific assumptions adopted in this study were outlined below:

- (1). The molten liquid TC4 titanium alloy was treated as an incompressible laminar flow.
- (2). The laser heat source model was modeled using the Gaussian rotating heat source model, and the radius of the beam is assumed to be constant.

- (3). The chemical interaction between the phases were neglected.
- (4). The individual thermal physical parameters of the material change linearly with temperature, such as density, specific heat capacity, viscosity, etc., and other properties are a fixed value, which varies between the solid–liquid phase.

Despite capturing the main physical phenomena of VLBW, these simplifications introduce limitations. Treating the molten metal as laminar flow neglects possible turbulence effects, and assuming a constant beam radius ignores potential focus fluctuations or variations in the beam profile. Ignoring chemical interactions and linearizing material properties may affect local predictions, while simplified boundary conditions for convection and radiation do not fully account for environmental disturbances. Consequently, although the model provides reliable qualitative insights into the process, caution should be exercised when interpreting quantitative results under extreme process conditions.

3.2. Governing equation

The molten pool flow in VLBW of titanium alloy adheres to the principles of physical conservation, where the fluid motion must satisfy the fundamental conservation laws. The simulation of molten pool flow is achieved by solving the governing equations that describe the dynamics of the system. The fundamental governing equations this flow are as follows[[30], [31], [32]]:

Continuity equation:

$$\nabla \bullet \vec{v} = 0 \quad (1)$$

where the v is the velocity vector. Mass conservation equation:

$$\frac{\partial \rho}{\partial t} = -\nabla \bullet (\rho \vec{v}) \quad (2)$$

where t is the time and ρ is the density.

Momentum conservation equation:

$$\frac{\partial(\rho\vec{v})}{\partial t} + \nabla \bullet (\rho\vec{v}\vec{v}) = -\nabla N + \nabla \bullet (\varphi \nabla \vec{v}) - M\vec{v} - \rho g(\kappa T - \kappa T_{ref} - 1) \quad (3)$$

where N is static pressure, g is gravitational acceleration, T and T_{ref} represent liquid phase temperature and reference temperature respectively, φ is viscosity, κ is thermal diffusion coefficient, M represents a function related to f_l , given by the following equation:

$$M = D \frac{(1-f_l)}{f_l+B} \quad (4)$$

where B is a decimal greater than zero, D is the paste region constant, and f_l is the volume fraction of the liquid phase, given by the following formula:

$$f_l = \begin{cases} 0, Q < Q_1 \\ \frac{Q-Q_1}{Q_2-Q_1}, Q_1 < Q < Q_2 \\ f_l, Q > Q_2 \end{cases} \quad (5)$$

where Q_1 is the liquidus temperature and Q_2 is the solid phase line temperature.

Energy conservation equation:

$$\frac{\partial(\rho Z)}{\partial t} + \nabla \bullet \rho v Z - \nabla \bullet (\zeta \nabla T) = S_E \quad (6)$$

where Z is the enthalpy of mixing, ζ is the thermal conductivity, and S_E represents the energy source term.

3.3. Driving force

In laser welding, the presence of the driving force is known to affect the flow of liquid metal within the molten pool, the behavior of the molten pool during welding, heat conduction, and ultimately the quality of the weld. As a result, it cannot be ignored. The

main driving force in this simulation is given as follows [33]:

$$\mathbf{F} = \mathbf{F}_r + \mathbf{F}_s + \mathbf{F}_b \quad (7)$$

where F_r is the recoil pressure of rapid solidification of liquid metal in molten pool; F_s is the surface tension generated by the liquid surface molecules; F_b is the buoyant force present in the liquid.

F_r can be described by the following formula:

$$F_r = IB_0T_s^{-\frac{1}{2}} \exp\left(-\frac{M_aL_v}{N_a\sigma T_s}\right) \quad (8)$$

where I is the coefficient related to environmental pressure; B_0 and T_s are the vaporization constant and surface temperature, respectively. M_a represents the relative atomic mass of the element. L_v represents the amount of heat absorbed by the liquid per unit mass during vaporization. N_a and σ represent the Avogadro constant and the Boltzmann constant, respectively.

F_s is represented by:

$$F_s = \kappa\left(\mu_0 + \frac{d\mu}{dT}(T - Q_L)\right) \quad (9)$$

where κ represents the unconstrained surface curvature; Q_L and μ_0 are the melting point and surface tension coefficient of metal, respectively. μ represents the surface tension.

F_b is given by the following formula:

$$F_b = \rho\beta(T - T_l)\bullet|g| \quad (10)$$

where ρ is the density; β is the coefficient of thermal expansion.

3.4. Heat source model

The heat source model adopted for VLBW is Gaussian surface heat source model and Gaussian rotating body heat source model.

The Gaussian surface heat source model is depicted as follows [34]:

$$q_s(x, y) = \frac{2Q}{\pi R_0^2 H_1} \exp\left(\frac{-3r^2}{R_0^2}\right) \quad (11)$$

where q_s is the Gaussian surface heat source effective energy, Q is the effective heat source power, R_0 is the beam radius, H_1 is the surface heat source penetration depth and r is the effective radius of the heat source.

The Gaussian rotating body source model is depicted as follows [35]:

$$q(x, y, z) = \frac{9Q}{0.95\pi R_0^2 H_2} \exp\left(\frac{-4r^2}{R_0^2 \log\left(\frac{H_2}{-z}\right)}\right) \quad (12)$$

where, Q is the effective heat source power; R_0 is the beam radius; H_2 is the body heat source penetration depth; r is the effective radius of the heat source.

3.5. Boundary conditions

The thermal boundary conditions on the keyhole and molten pool free surfaces, which are composed of laser energy input, radiative losses, and evaporation-induced heat loss, can be described as follows:

$$-k \frac{\partial T}{\partial n} = q_{laser} - q_{rad} - q_{evap} \quad (13)$$

where q_{laser} represents the laser energy input per unit area, q_{rad} represents the radiative heat loss per unit area, q_{evap} denotes the energy loss due to evaporation per unit area.

$$q_{laser} = q_s(x, y) + q(x, y, z) \quad (14)$$

$$q_{rad} = \epsilon \sigma (T^4 - T_{env}^4) \quad (15)$$

where ϵ representing the material's emissivity, σ is the Stefan–Boltzmann constant, T is the surface temperature, T_{env} is the ambient temperature.

$$q_{evap} = \dot{m}_{evap} \cdot L_v \quad (16)$$

where $\dot{m}_{evap}$ is the mass flux of evaporation per unit area, L_v is the latent heat of evaporation.

The evaporation flux can be approximated using the temperature-based Knudsen equation:

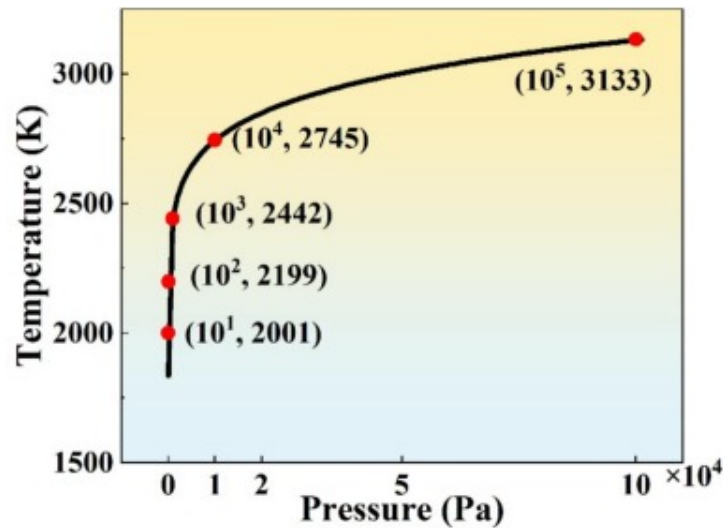
$$\dot{m}_{evap} = \alpha \cdot P_{sat}(T) \cdot \sqrt{\frac{M}{2\pi RT}} \quad (17)$$

where α is the evaporation coefficient, $P_{sat}(T)$ is the saturation vapor pressure of the metal surface, M is the molar mass of the vapor molecules, and R is the universal gas constant.

3.6. Numerical implementation

In this study, CFD software was employed to discretize the entire spatial domain. A simplified welding region with dimensions of 10mm×30mm×60mm is selected for numerical modeling, as illustrated in Fig. 5(a). In order to better simulate the vacuum environment, based on the study of empirical data, a plasma of TC4 titanium alloy with extremely low density was introduced into the gas zone on both sides of the model. Fig. 4 shows the line graph of the boiling point of TC4 titanium alloy varying with pressure, calculated by the Clausius-Clapeyron equation [36]:

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H_{vap}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad (18)$$



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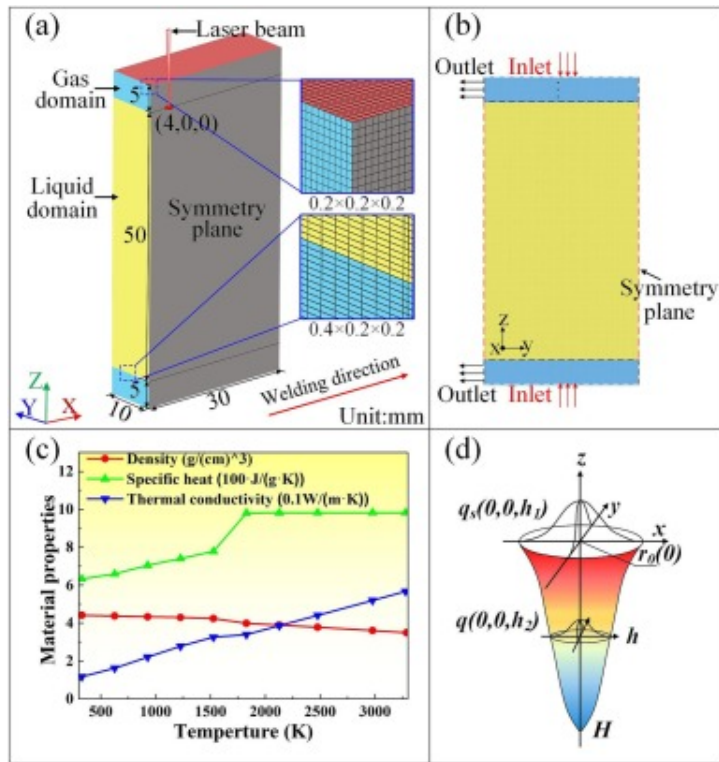
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Fig. 4. The boiling point of TC4 titanium alloy under various ambient pressure.

where the P_1 and P_2 are two different pressures; T_1 , T_2 are the boiling points at the corresponding pressures; ΔH_{vap} is the enthalpy of vaporization; R is the gas constant.

We have established half of the model. Taking the adiabatic surface in the middle as the interface, the model is divided into a total of 1,575,000 grid elements to save computing time while ensuring accuracy. The grid size of the near-weld area was refined to $0.2\text{mm} \times 0.2\text{mm} \times 0.2\text{mm}$, and the grid size of the weld area was further reduced to $0.4\text{mm} \times 0.2\text{mm} \times 0.2\text{mm}$, ensuring the high resolution of the simulation and the accuracy of the analysis. Fig. 5(b) depicts the boundary conditions from the YZ plane perspective, with the pressure inlet and pressure outlet positioned in the top and bottom gas domains, respectively. During the welding process, the temperature exerted a considerable influence on the material properties of the TC4 titanium alloy. To account for this, temperature-dependent material property curves of Ti-6Al-4V, calculated using JMatPro software based on the elemental composition, were integrated into the simulation. These

curves precisely captured variations in key properties such as density, specific heat capacity, and thermal expansion coefficient across different temperature ranges, as illustrated in [Fig. 5\(c\)](#). According to relevant studies [37,38], the significant reduction of ambient pressure in VLBW effectively suppresses the expansion of plasma during the welding process and substantially mitigates the recoil pressure generated by metal vapor. Compared to atmospheric conditions, this phenomenon leads to a remarkable increase in weld penetration depth. To accurately characterize the thermodynamic behavior of the laser heat source on the material surface, this study adopted both Gaussian surface heat source and Gaussian rotational body heat source model to systematically describe the spatial distribution characteristics of laser energy density. The schematic illustration of the heat source model is shown in [Fig. 5\(d\)](#). The thermophysical properties of materials were critical factors influencing heat transfer and welding performance. The key thermophysical parameters of Ti-6Al-4V titanium alloy were comprehensively listed in [Table 3](#).



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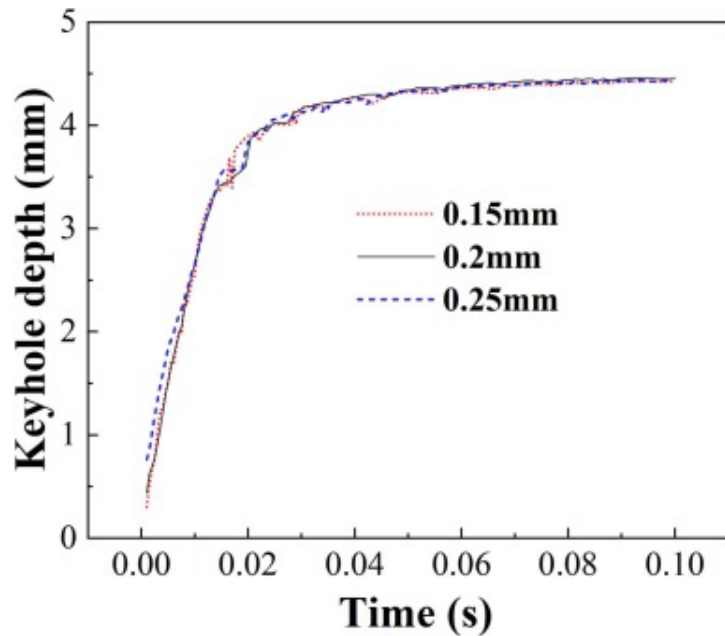
Fig. 5. The mesh model of computational domain and boundary conditions: (a) Over all 3D map; (b) Y-Z plane; (c) The relevant thermophysical parameters; (d) Combination of Gaussian surface and body heat source models.

Table 3. Thermophysical properties of Ti-6Al-4V titanium alloy [39].

Name	Symbol	Value	Units
Solidus temperature	T_s	1878	K
Liquidus temperature	T_l	1923	K

Name	Symbol	Value	Units
Vaporization temperature	T_v	3133	K
Radiation emissivity	ϵ	0.3	/
Latent heat of fusion	L_m	2.86×10^5	J
Latent heat of evaporation	L_v	4.19×10^6	J
Thermal expansion coefficient	β	8.8×10^{-6}	K^{-1}
Surface tension	γ_0	1.2	$\text{N} \cdot \text{m}^{-1}$
Surface tension gradient	$d\gamma/dT$	-0.0002	$\text{N} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$

To evaluate the influence of mesh resolution on the numerical results, a grid independence test was conducted. The reference mesh size of 0.2mm was compared with finer (0.15mm) and coarser (0.3mm) meshes, and the keyhole depth was selected as the representative evaluation parameter. The maximum keyhole depths obtained under the three mesh sizes were 44.29mm, 44.6mm, and 44.4mm, respectively. As illustrated in Fig. 6, the three curves exhibit close agreement, and the variation in maximum depth is less than 1%. This indicates that the numerical predictions are insensitive to further refinement, and thus the 0.2mm grid was adopted in subsequent simulations as a balance between computational accuracy and efficiency.



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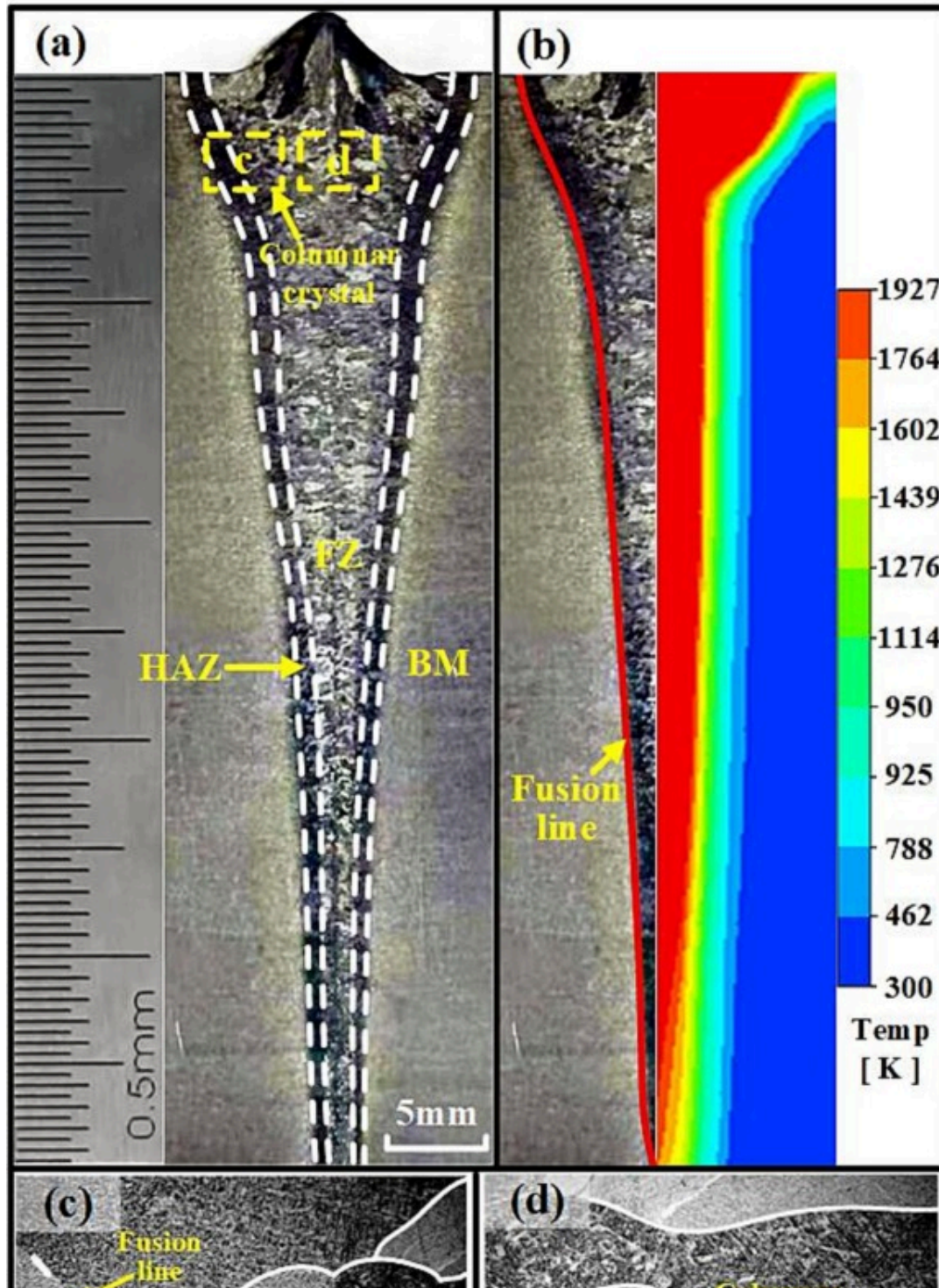
Fig. 6. Grid independence test: variation of keyhole depth with time under different grid sizes.

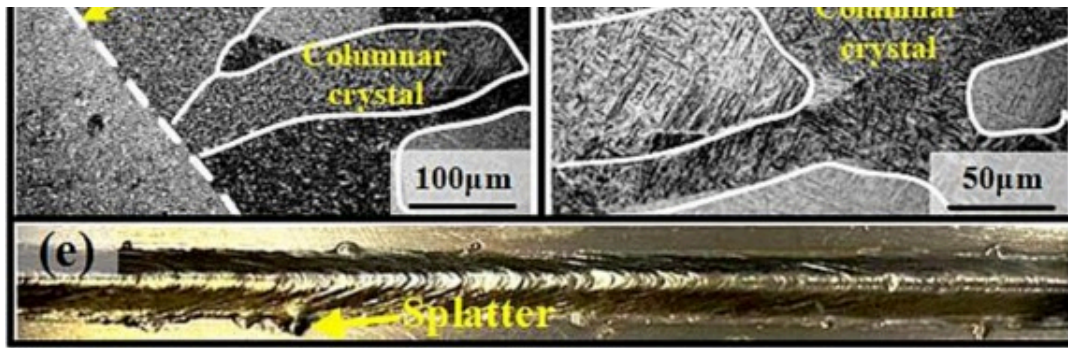
4. Results and discussion

4.1. Macroscopic morphology analysis and model verification

Fig. 7(a) illustrates the macrostructure of the cross-section of the TC4 titanium alloy weld produced via VLBW. The weld, measuring 50mm in length, exhibits a slender, nail-like geometry with a wider top section and a narrower bottom section, highlighting the characteristic “deep penetration” welding behavior under vacuum conditions. Notably, the cross-section shows no microstructural defects such as porosity, undercut, or lack of fusion, demonstrating high-quality weld formation. In the FZ regions, elongated columnar crystal structures are distinctly visible, growing perpendicularly from the fusion

boundary toward the center. The magnified views of these columnar crystals are shown in [Fig. 7\(c\)](#) and [Fig. 7\(d\)](#). This growth pattern adheres to the principle of minimum energy, following the direction of the steepest temperature gradient.





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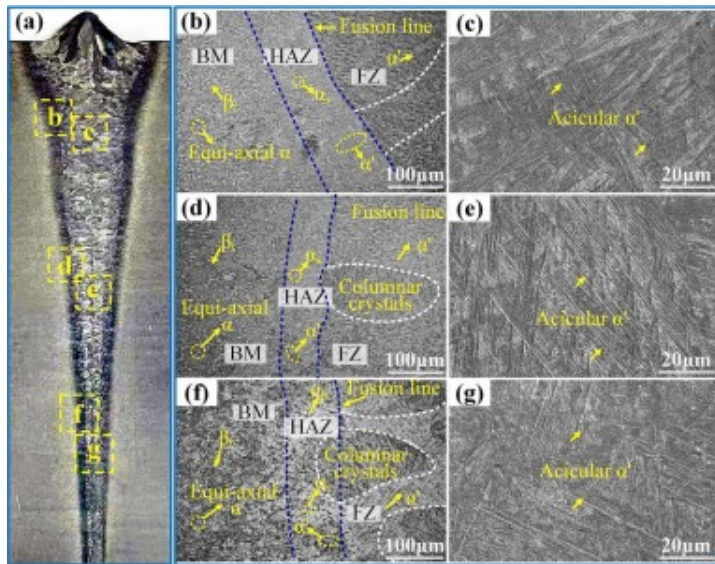
Fig. 7. Macrograph of the TC4 titanium alloy weld seam and model validation during VLBW: (a) Microstructure diagram of the weld cross-section; (b) Model verification; (c) Magnified view of position c in (a); (d) Magnified view of position d in (a); (e) Fish scale bead diagram.

To validate the accuracy of the established model, the simulated fusion lines of the weld cross-section were compared with experimental results, as shown in [Fig. 7\(b\)](#). The close agreement between the simulated and experimental data confirms the reliability of the numerical model. Additionally, the surface morphology of the weld displays a uniform, fish-scale pattern, indicative of process stability despite minor spattering during welding, as depicted in [Fig. 7\(e\)](#).

4.2. Microstructure analysis

[Fig. 8](#) illustrates the microstructure distribution of 50mm-thick TC4 titanium alloy laser-welded joints under vacuum conditions. Due to the vacuum environment, the weld exhibits deep penetration characteristics, with significant variations in heat input along the penetration depth [21]. The microstructure of the weld zone in different regions, as shown in [Fig. 8](#), [Fig. 8](#), [Fig. 8](#), was analyzed using OM. The weld zone is divided into three distinct regions: the BM, the HAZ and the FZ. In the BM region, the microstructure

consists of equiaxed primary α (α_p) phase and intergranular β phase. The HAZ region, which is relatively narrow, is characterized by a combination of incompletely transformed coarse α_p phase, fine-lamellar secondary α (α_s) precipitated during cooling from the high-temperature β phase field, and a minor fraction of fine acicular α' phase resulting from localized rapid cooling [40]. The formation of α_s is attributed to a diffusion-controlled phase transformation under moderate cooling rates, whereas the limited α' martensite originates from shear-dominated transformation in regions experiencing higher cooling rates. In the FZ region, the microstructure undergoes nearly complete transformation due to melting and rapid solidification. The extreme cooling rate inherent to laser welding promotes a diffusionless martensitic transformation, resulting in a microstructure predominantly composed of fine acicular α' phase. The microstructure of the FZ in the TR, MR and BR observed by SEM is presented in Fig. 8, Fig. 8, Fig. 8. The FZ is predominantly composed of orthogonally arranged acicular α' phase, suggesting rapid cooling across the entire thickness direction and significant non-equilibrium phase transformations. This microstructure originates from the prior β phase field. During solidification, the molten metal first crystallizes into columnar grains of β phase, which grow epitaxially from the partially melted BM along the maximum temperature gradient. During subsequent cooling, the hexagonal α' martensite phase nucleates at the boundaries of these prior β columnar grains and grows inward, forming parallel α' phase structures that penetrate the entire β phase columnar grains. The transformed martensitic structures typically grow along specific crystallographic planes, forming a basket-weave structure within the confines of each prior β grain [40,41]. Within each basket-weave structure, the α' grains exhibit uniform orientation, adhering to the Burger's orientation relationship $\{0001\}\alpha' // \{110\}\beta$ and $\langle 11-20 \rangle \alpha' // \langle 111 \rangle \beta$ [42,43], while no preferential orientation is observed between different basket-weave structures due to the random orientation of the prior β grains.



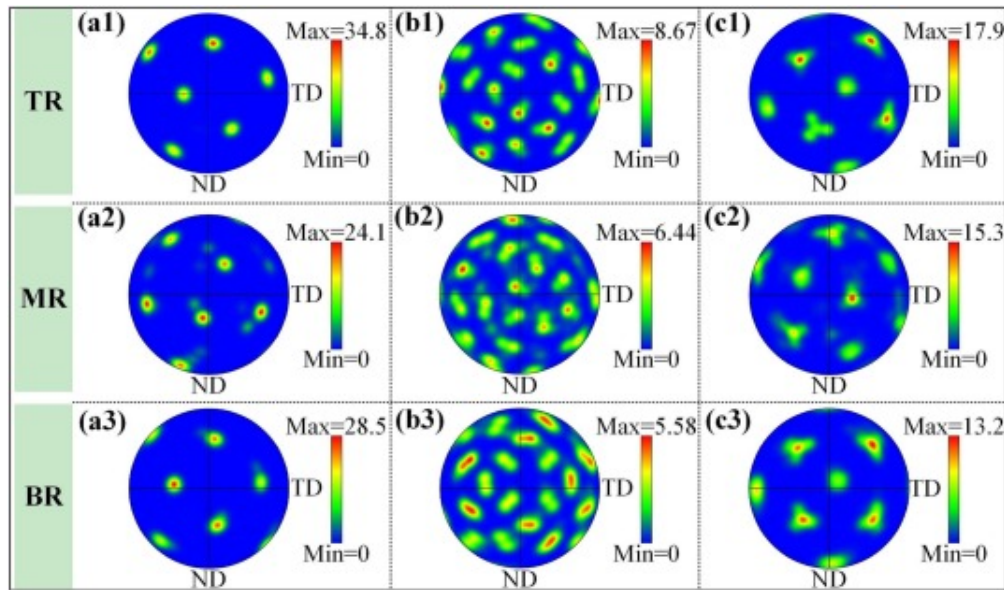
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Fig. 8. The microstructure distribution of the 50mm thick VLBW joint: (a) overall morphology of the weld joint cross-section; (b) and (c) TR of the joint; (d) and (e) MR of the joint; (f) and (g) BR of the joint.

The pole figures (PFs) for TR, MR and BR of FZ are shown in Fig. 9, it depicted the distribution and intensity of grain orientations in the FZ. The α -phase in the TR, MR and BR primarily aligns along the $\{0001\}$ and $\{11\bar{2}0\}$ planes. The strongest texture was observed in the TR with maximum values of 34.8 and 17.9, respectively. During the VLBW process of thick plates, the texture strength of the weld seam from top to bottom shows a decreasing trend. The temperature gradient in the TR is the greatest, and the heat flow direction is stable, which is conducive to the preferentially oriented growth of columnar crystals along the heat flow direction and the formation of a stronger texture. In contrast, the temperature gradient at the bottom of the weld seam is significantly reduced, the cooling rate is accelerated, and the texture strength is significantly weakened. In addition, the TR may also be subject to the repeated thermal cycling effect of the laser, which is

conductive to the further growth of the preferred grains and the strengthening of the texture.



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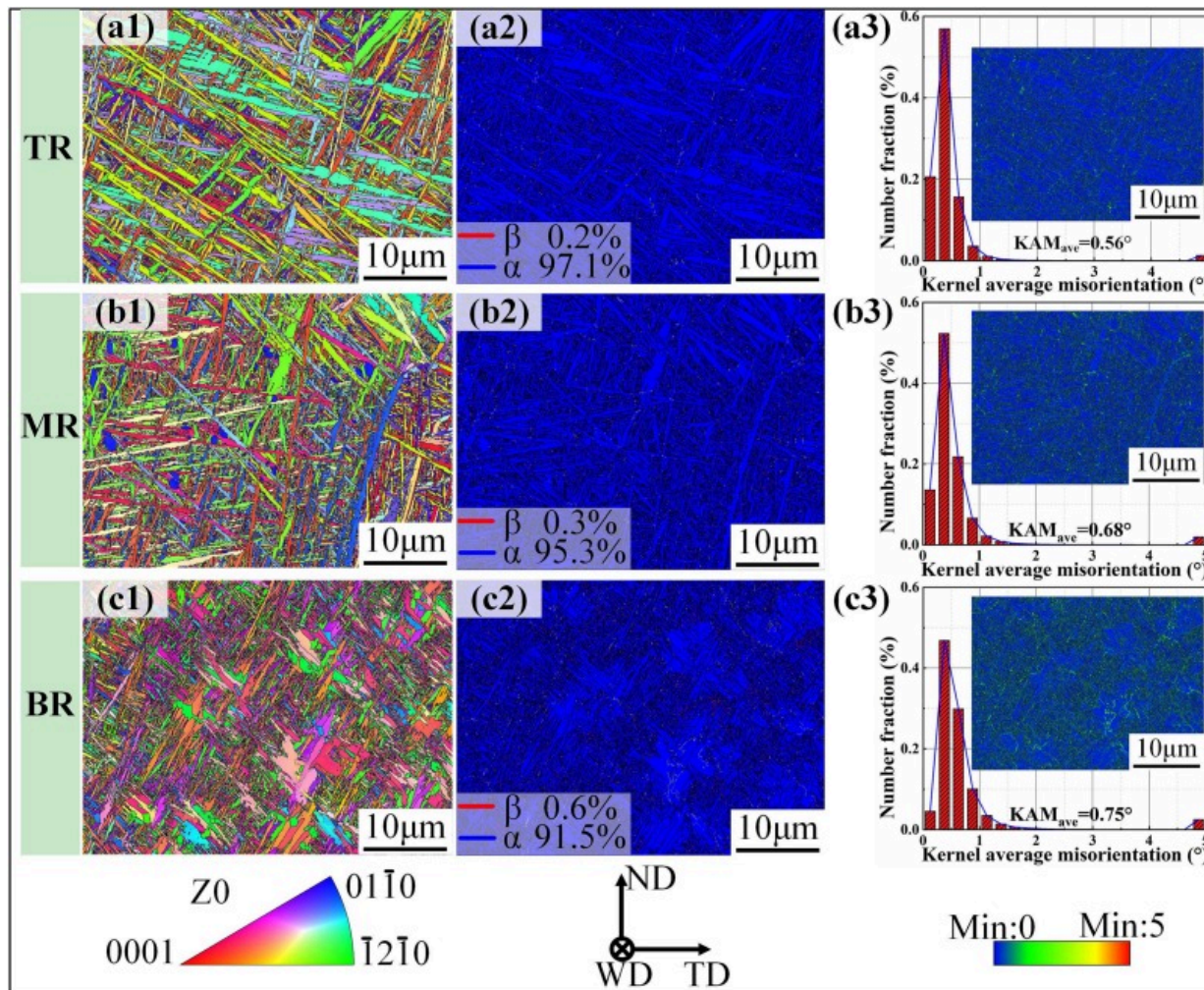
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Fig. 9. The pole figures (PFs) of each region in the FZ: (a1)-(a3) {0001} plane; (b1)-(b3) {10-10} plane; (c1)-(c3) {11-20} plane.

During the VLBW of TC4 titanium alloy, the substantial material thickness leads to significant variations in temperature gradients, heat input, and cooling rates, significantly influencing the microstructure and texture development across the TR, MR and BR. Additionally, increased material thickness introduces greater complexity to the weld pool flow, which will be further explored in subsequent simulations.

The grain orientation, phase distribution and Kernel Average Misorientation (KAM) diagrams in the TR, MR And BR of FZ in the VLBW welded joint are shown in Fig. 10. It can be seen from Fig. 10(a1) – (c1) that the microstructure of the TR, MR and BR is mainly composed of needle-like α 'martensite, which benefits from the relatively large cooling

rate in the weld area. These needle-like structures interweave with each other to form a basket weave pattern, and there is no obvious preferred orientation in the view area. The phase distribution in the FZ is shown in Fig. 10(a2)–(c2). All three regions are composed of α and β phases, with the volume fraction of each phase indicated. During rapid cooling, a large portion of the β phase undergoes a shear-induced transformation into the α phase, while a small amount of retained β phase maintains its original morphology, resulting in a microstructure predominated by the α phase with a minor fraction of β phase. Since the β phase constitutes less than 1%, it is difficult to observe clearly in the microstructure. From the TR to the BR, the content of the β phase gradually decreases while that of the α phase increases, which is attributed to the higher cooling rate in the BR. Fig. 10(a3) – (c3) shows the KAM graph and its statistical histogram, reflecting the distribution of residual strain. The green area indicates obvious local plastic deformation and stress concentration, indicating the regions where the grains are more prone to plastic deformation. The average KAM values of TR, MR and BR were 0.56° , 0.68° and 0.75° respectively, gradually increasing along the direction of the welding depth. The KAM value at the bottom of the weld seam is the highest, indicating that the intracrystalline plastic strain and dislocation density are the greatest. This is because the cooling rate at the bottom is the highest, resulting in a stronger thermal stress gradient and plastic deformation. Meanwhile, the small grains formed by rapid solidification may have greater intra-grain mismatch and orientation distortion. Moreover, the grain size gradually decreases from top to bottom. Smaller grains mean more grain boundaries, and the intragranular strain may also be more concentrated.



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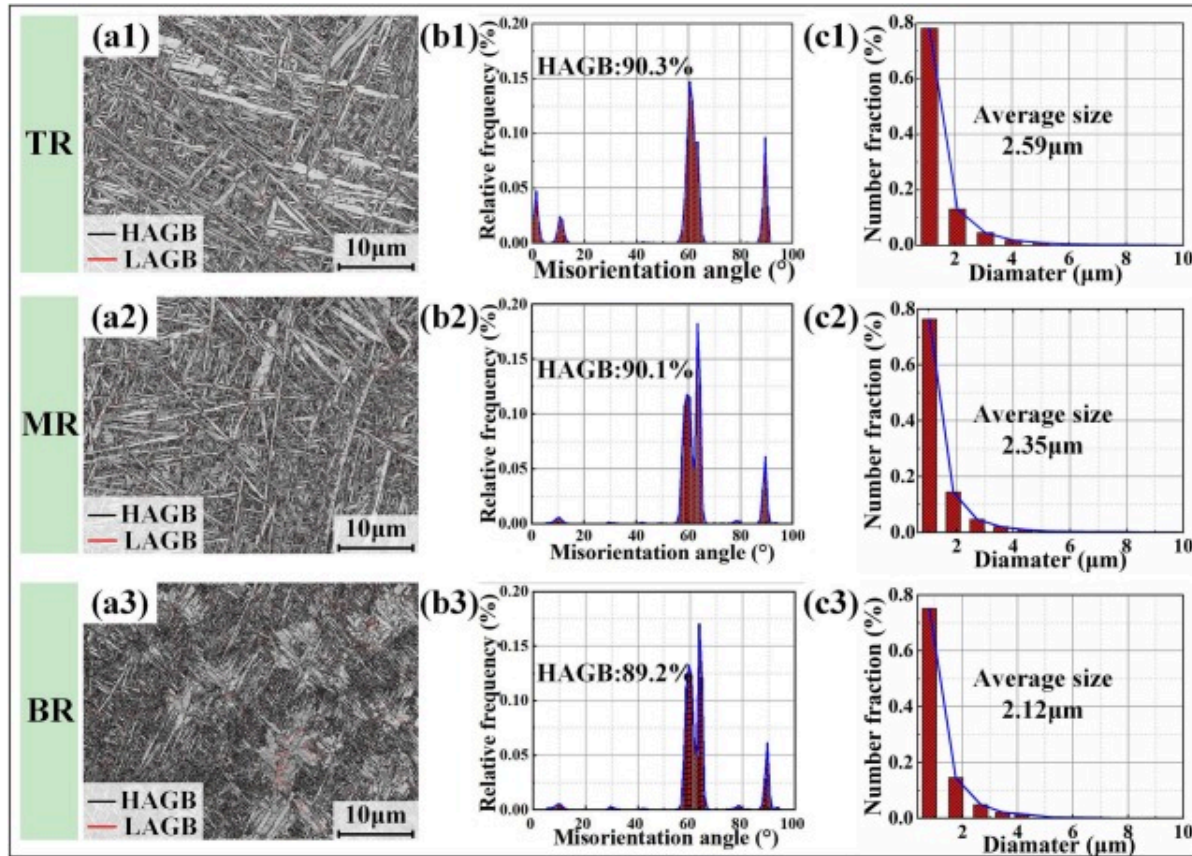
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Fig. 10. EBSD analysis of the FZ in each region: (a1)-(c1) Inverse pole figure (IPF) orientation map of TR, MR and BR; (a2)-(c2) Phase distribution of TR, MR and BR; (a3)-(c3) KAM maps and the statistics histogram of TR, MR and BR.

The distribution of grain boundary in the TR, MR and BR of FZ region is depicted in Fig. 11 (a1)-(a3), with HAGB being predominant across all regions. The grain orientation

difference angles and HAGB statistics for these regions are presented in Fig. 11(b1)-(b3).

The orientation difference angles of α' martensite in all three regions are primarily concentrated around 60° . The average proportions of HAGBs ($>15^\circ$) in the TR, MR and BR was nearly identical, at 90.3%, 90.1%, and 89.2%, respectively. The rapid cooling rate in the FZ significantly influences the orientation difference distribution. During cooling, the acicular martensite follows the Burgers orientation relationship, and the presence of additional α'/α' interfaces in the TR further increases the proportion of HAGBs.



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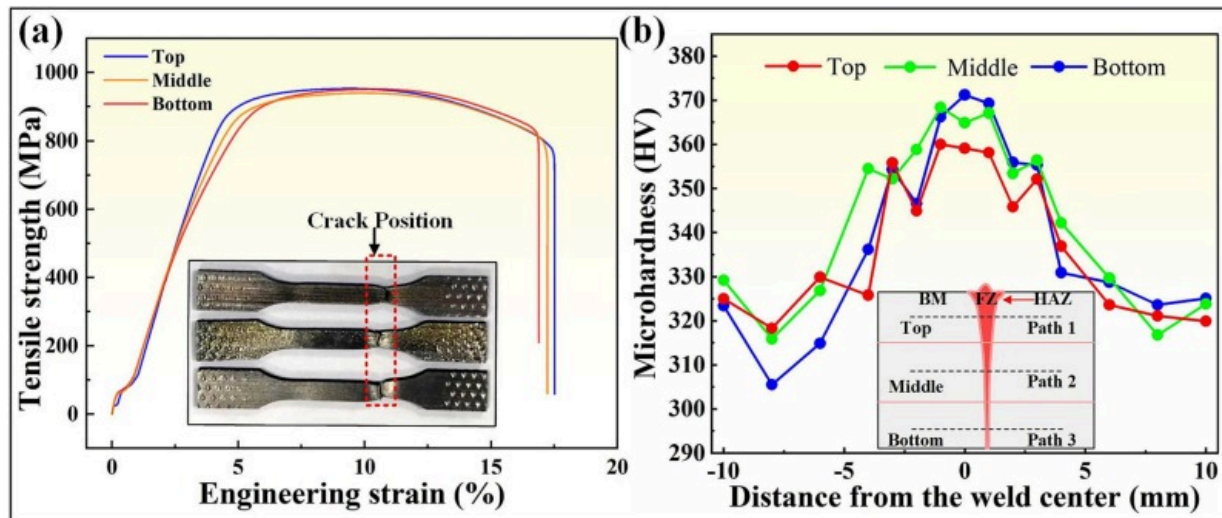
Fig. 11. Characterization of grain boundaries, misorientation angles, and grain sizes in different regions: (a1)-(a3) High angle and low angle grain boundaries of TR, MR and BR;

(b1)-(b3) Misorientation angle of TR, MR and BR; (c1)-(c3) Statistical graph of grain size of TR, MR and BR.

The statistical analysis of the average grain size in the TR, MR and BR of FZ are presented in Fig. 11(c1)-(c3). The grain size in the FZ predominantly falls within the range of 0 – 5 μ m. The average grain sizes of TR, MR and BR were 2.59 μ m, 2.35 μ m and 2.12 μ m respectively, gradually decreasing from TR to BR. The most important factor determining the grain size is the cooling rate. The greater the cooling rate, the finer the grains. For thick plates under vacuum conditions, TR is close to the laser heat source, receives the maximum heat, has a relatively low cooling rate, and the grains have sufficient time to grow. In contrast, the MR and BR areas receive less heat, and the bottom is in contact with the processing platform, conducting heat more quickly and forming a larger cooling rate. In addition, the suppressed thermal convection under vacuum conditions increases the undercooling and nucleation rates at the solidification front. Therefore, from TR to BR, as the cooling rate increases, the grain size gradually decreases. Ultimately, finer grains are produced in these areas.

4.3. Mechanical properties analysis

The tensile stress–strain curves of the VLBW specimens are shown in Fig. 12(a). Fracture consistently occurred in the BM for samples taken from all three FZ regions, indicating the higher strength of the FZ. The tensile strengths of TR, MR, and BR were 947MPa, 940MPa, and 951MPa, respectively, which are very close to each other, with BR showing the highest value, likely due to its finer grain size. The stress–strain curves of TR, MR, and BR exhibited similar shapes without distinct yield points, which may be attributed to the formation of martensitic hardening phases during rapid cooling. Fig. 12(b) presents the microhardness distributions of TR, MR, and BR at different positions across the FZ. A saddle-shaped profile is observed along the thickness direction. The maximum hardness in the BM is 330 HV, which is notably lower than that in the HAZ and FZ of BR, reaching 358 HV and 371 HV, respectively. This hardness increase is attributed to grain refinement and the formation of dense α'/α' interface structures during rapid solidification.



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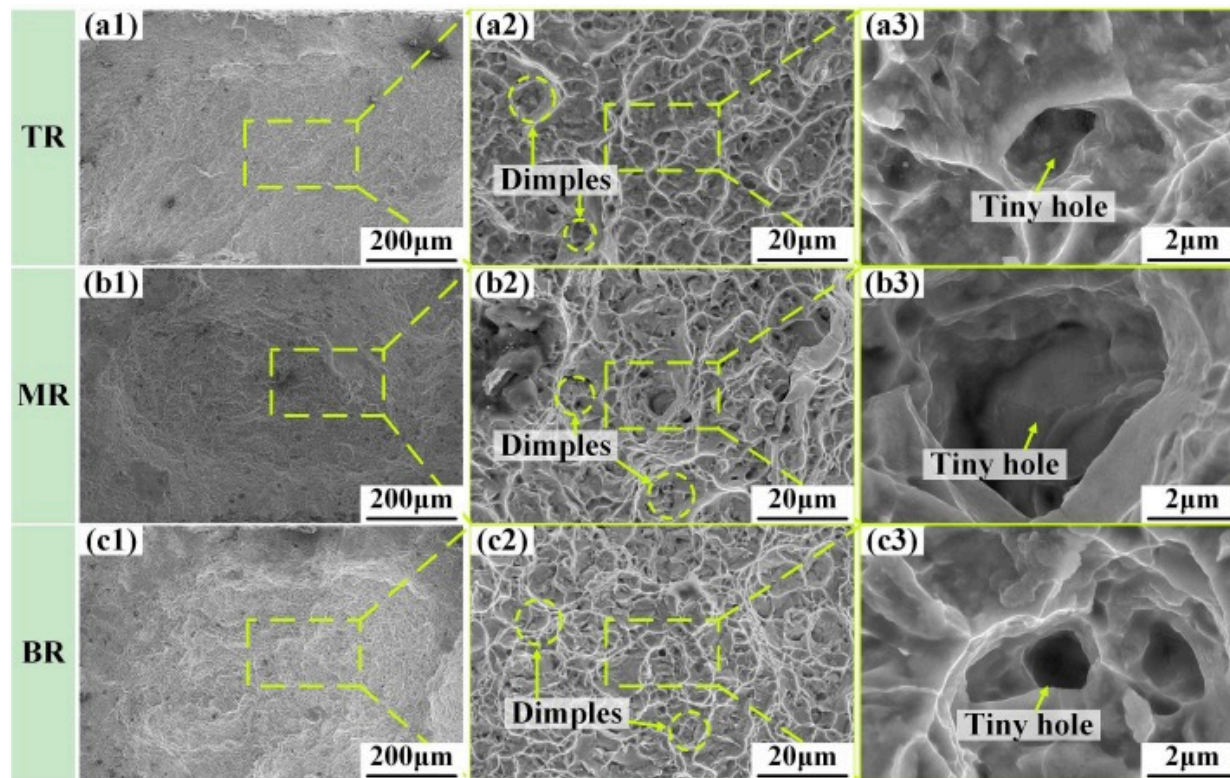
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Fig. 12. Mechanical properties of TC4 titanium alloy in different regions under VLBW: (a) Stress–strain curve and fracture position of tensile specimen in TR, MR and BR; (b) Transverse microhardness in TR, MR and BR of TC4 titanium alloy welded joint by VLBW.

Additionally, the hardness values of BR and MR are generally higher than those of TR at the same locations. This is mainly due to the decreasing grain size from top to bottom and the higher cooling rate in the lower regions, which promotes the formation of more α' martensite and α'/α' interfaces. In contrast, the TR undergoes slower cooling and receives more heat input, leading to grain coarsening. These results confirm that the hardness at the top of the FZ is lower than that at the bottom.

The fracture microstructure of tensile specimens in the TR, MR and BR is shown in Fig. 13. It can be observed that a large number of dimples are present in all three regions, and the fractures appear dark gray and fibrous, characteristic of ductile fracture. These dimples are microscopic voids formed during the plastic deformation process at the micro-area, which are nucleated, grown, and aggregated, and eventually coalesce to cause

fracture, leaving distinct traces on the fracture surface. At the 200 μm scale, the fracture characteristics were observed to be relatively uniform across the specimen surfaces. Upon zooming in to the 20 μm scale, a significant number of dimples were identified in all regions, indicating a ductile fracture mode characterized by dimple coalescence. The shallow depth of these dimples suggests that the stress distribution in these areas was relatively uniform, with limited strain concentration. Further magnification to the 2 μm scale revealed the presence of tiny hole alongside localized cleavage planes, which points to a higher stress concentration in this specific region.



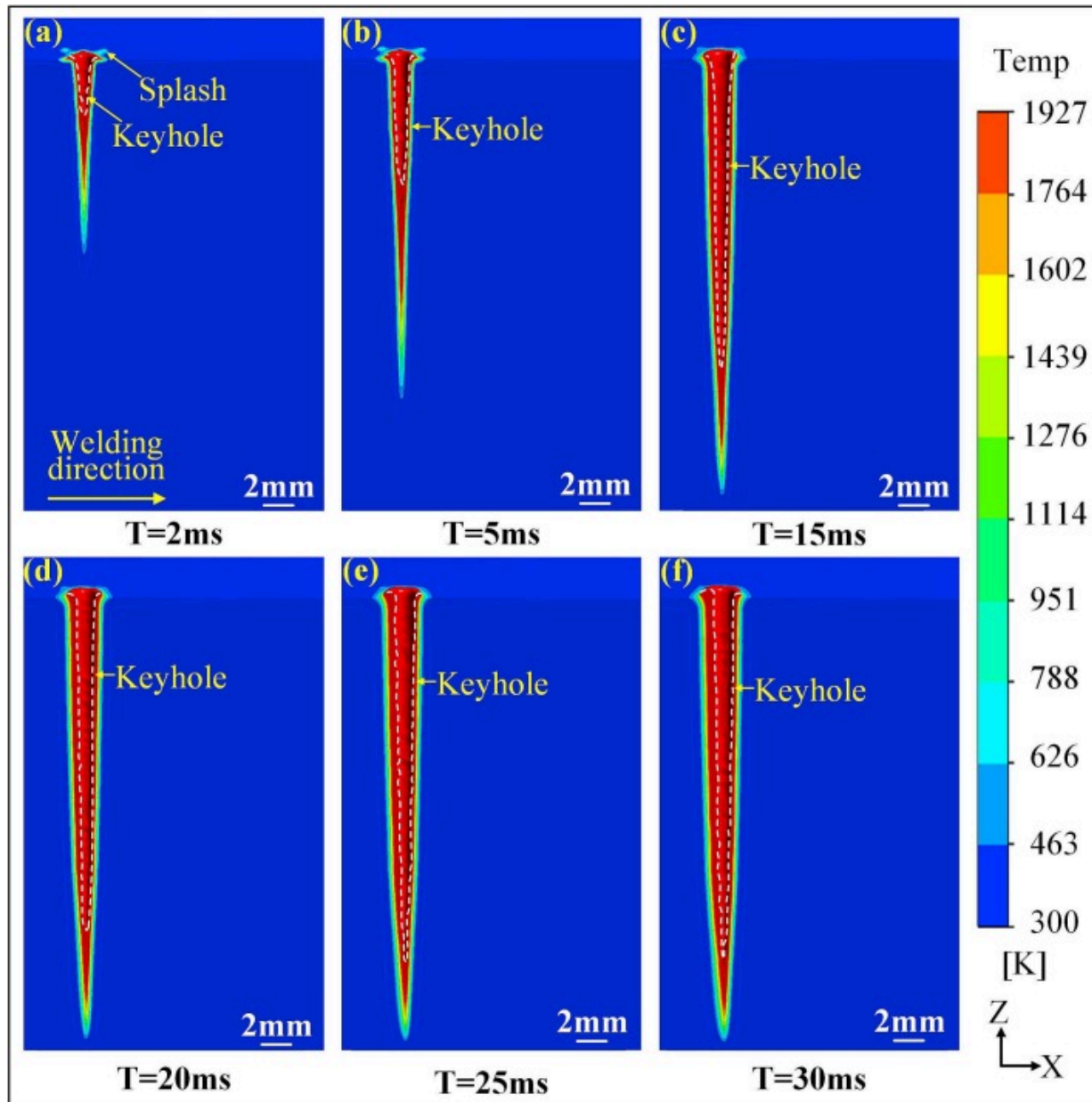
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Fig. 13. Fracture morphologies of TC4 titanium alloy tensile samples at different regions: (a1)-(a3) TR; (b1)-(b3) MR; (c1)-(c3) BR.

4.4. Characteristics of the thermo-flow field of the molten pool in VLBW

Fig. 14 illustrates the temporal evolution of the keyhole along the XZ plane. In the initial stage, as shown in Fig. 14(a)–(b), the keyhole begins to form, accompanied by a small amount of spatter near the TR of the molten pool. As the process progresses, Fig. 14(c)–(d) shows that by 20ms, the region influenced by the laser beam extends to the bottom of the material; however, the keyhole continues to grow in depth. Fig. 14(e)–(f) depicts the stage at which the keyhole becomes fully developed and reaches a quasi-steady state. The keyhole attains its maximum depth at approximately 25ms, after which the molten pool continues to exhibit a gradual increase in temperature, indicating sustained energy input and thermal accumulation.



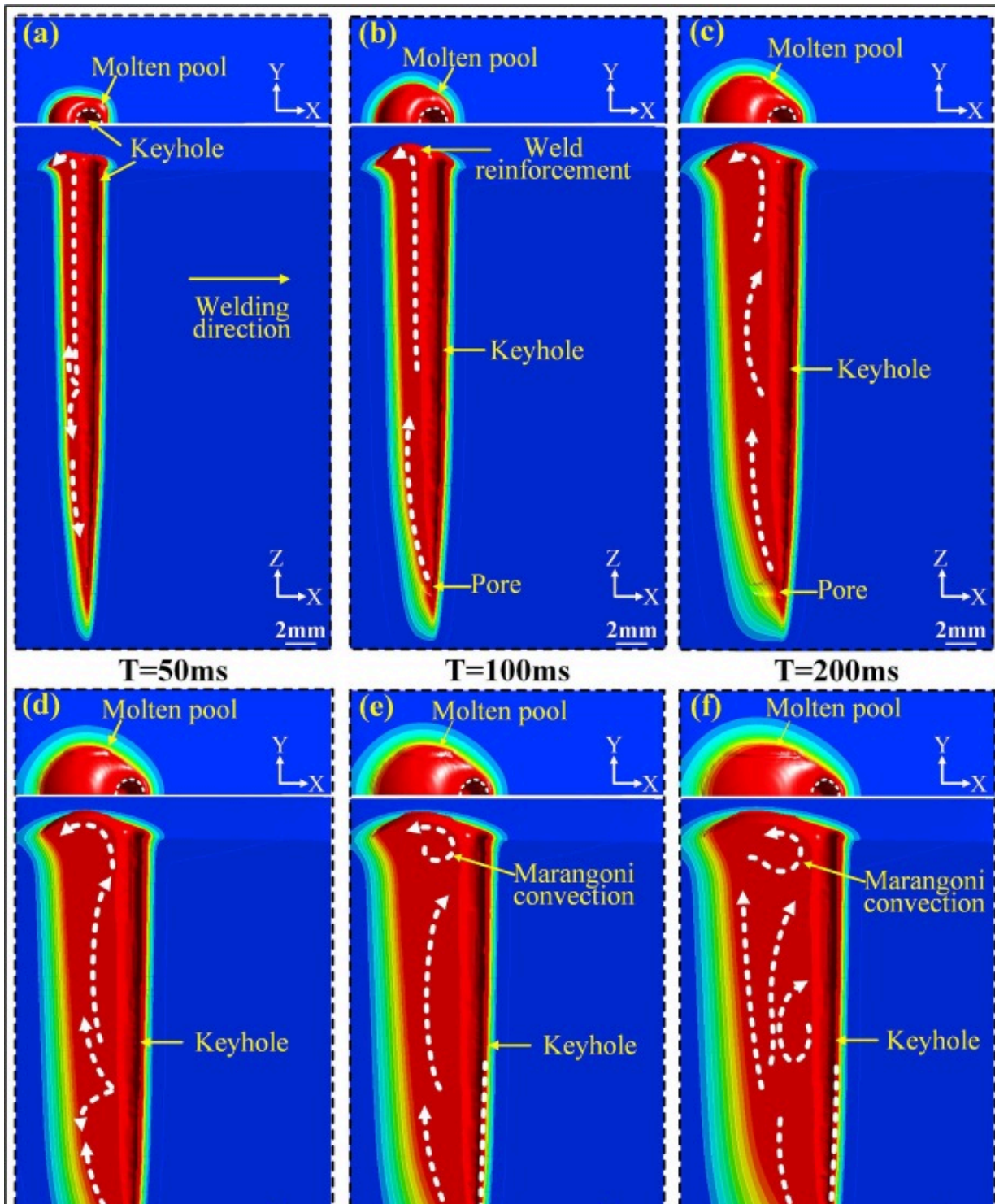
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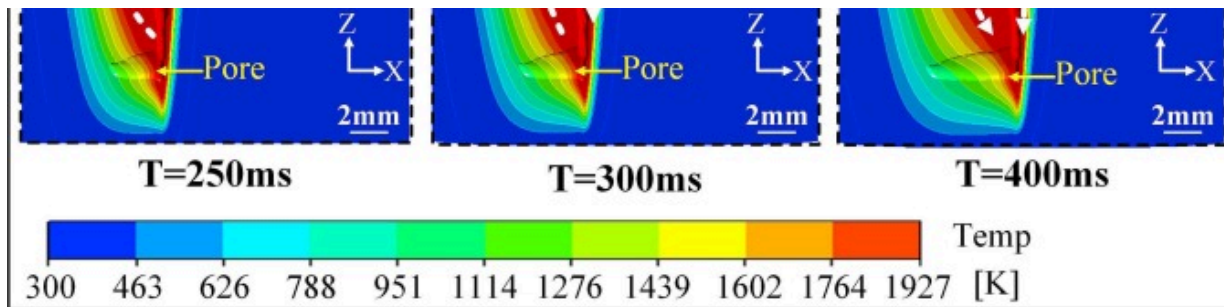
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Fig. 14. Longitudinal section of the keyhole evolution during VLBW process: (a) $t=2\text{ms}$; (b) $t=5\text{ms}$; (c) $t=15\text{ms}$; (d) $t=20\text{ms}$; (e) $t=25\text{ms}$; (f) $t=30\text{ms}$.

Fig. 15 shows the temporal evolution of the surface temperature distribution and internal flow dynamics of the molten pool after the stable formation of the keyhole in the VLBW process. As shown in Fig. 15(a), at 50ms, the molten pool is still very small and in a developing state. Under the combined action of recoil pressure and surface tension, the molten metal flows backward after the keyhole is locked. Among them, the negative dy/dT of TC4 titanium alloy indicates that the surface tension decreases with the increase of temperature. Therefore, the molten metal on the surface of the molten pool flows from the area with a lower temperature (higher surface tension) to the area with a higher temperature (lower surface tension), forming Marangoni convection and affecting the overall dynamics of the molten pool. When it reaches the rear wall of the molten pool, this flow is blocked and flows to the upper right under the action of thermal buoyancy. Due to the vacuum environment and the absence of ambient gas pressure, the natural heat convection loss is reduced, the backflow pressure effect is enhanced, and the reflux of the molten metal after the keyhole is intensified. As the heat source continues to move along the X direction, the molten pool keeps expanding forward, and the surface temperature field gradually increases (as shown in Fig. 15 (b)-(f)). When it reached 250ms, the width of the molten pool was close to a stable state, and the subsequent changes were relatively small. Fig. 15(b) shows the air hole formed at the bottom of the keyhole at approximately 100ms. The main reason is that under the drive of the strong backflushing pressure in the vacuum environment, the molten metal at the bottom of the keyhole flows to the rear wall. When the fluid encounters the side wall, it is forced to deflect upwards and to the right, causing a local collapse of the rear keyhole wall and subsequently forming a pore. As shown in Fig. 15(b)-(f), the hole remains connected to the keyhole throughout the process. However, the liquid metal at the bottom of the keyhole flows backward and downward, hindering the gas from escaping upward through the keyhole, causing the keyhole to gradually expand and eventually forming a distinct slender tail-like structure beneath and behind the molten pool. As can be seen from Fig.

15(e)-(f), the molten metal in the middle and upper areas of the molten pool flows counterclockwise along the keyhole wall. Under the action of Marangoni convection induced by surface tension and thermal buoyancy, a distinct vortex structure is formed. This vortex greatly alters the flow pattern and temperature distribution inside the molten pool.





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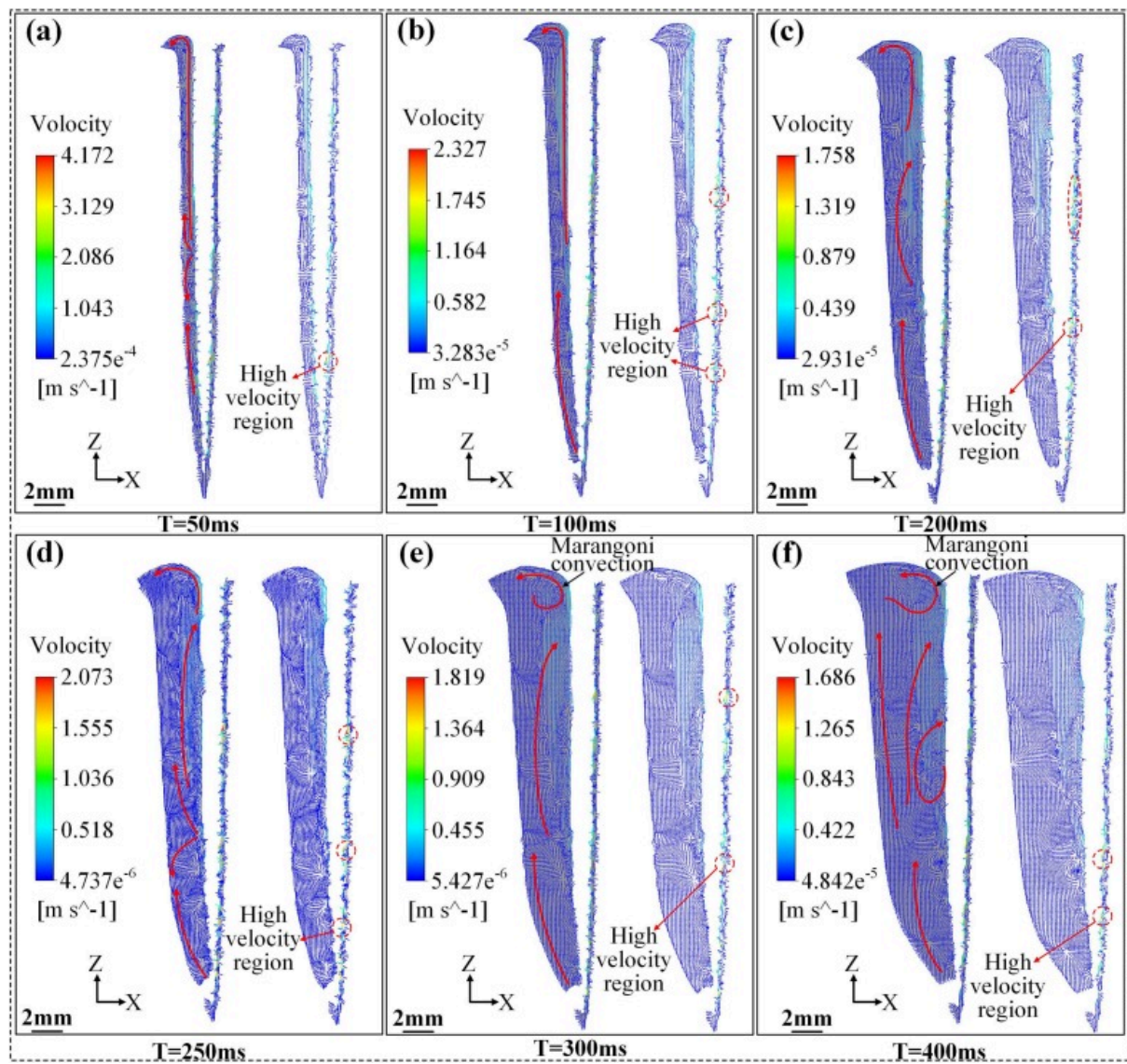
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Fig. 15. Flow pattern and surface deformation of the molten pool: (a) $t=50\text{ms}$; (b) $t=100\text{ms}$; (c) $t=200\text{ms}$; (d) $t=250\text{ms}$; (e) $t=300\text{ms}$; (f) $t=400\text{ms}$.

Although such small pores appear in the simulation, no pores were observed in the experimental cross-sections. This is consistent with previous reports that vacuum laser welding significantly reduces pore formation due to the suppression of keyhole instability and minimized gas entrapment[25,27]. Therefore, while minor pore formation can occur under extreme local flow conditions in the model, the overall vacuum-mediated keyhole dynamics effectively inhibit gas porosity, resulting in dense and high-quality welds. The experimental observations thus provide strong support for the simulation results and reinforce the understanding of keyhole flow mechanisms in vacuum laser welding.

Fig. 16 shows the fluid flow in the longitudinal section of the VLBW TC4 thick plate molten pool at different times. It can be known from Fig. 16 (a)-(f) that the maximum flow velocity in the molten pool is distributed on the walls before and after the keyhole. This is because there is no convective heat loss in a vacuum, resulting in stronger heat accumulation near the keyhole and a greater temperature gradient on the surface of the molten pool. These gradients enhanced the Marangoni effect. Therefore, under the combined drive of recoil pressure and the Marangoni effect, the liquid flow rate at the walls on both sides of the keyhole is relatively high. Meanwhile, with the movement of the heat source and the expansion of the molten pool, the maximum flow rate shows an

overall decreasing trend. This might be due to the fact that as the heat source moves, the temperature field gradually stabilizes, the temperature difference tends to balance, and the Marangoni effect weakens. From 300ms to 400ms ([Fig. 16\(e\)-\(f\)](#)), along the pool boundary, especially at the top and the front wall of the keyhole, obvious clockwise vortex structures can be observed. Under vacuum, the generation of these vortices is mainly driven by thermal buoyancy and the Marangoni effect. These vortices extend as the molten pool expands. Further affect the heat and mass distribution within the molten pool.



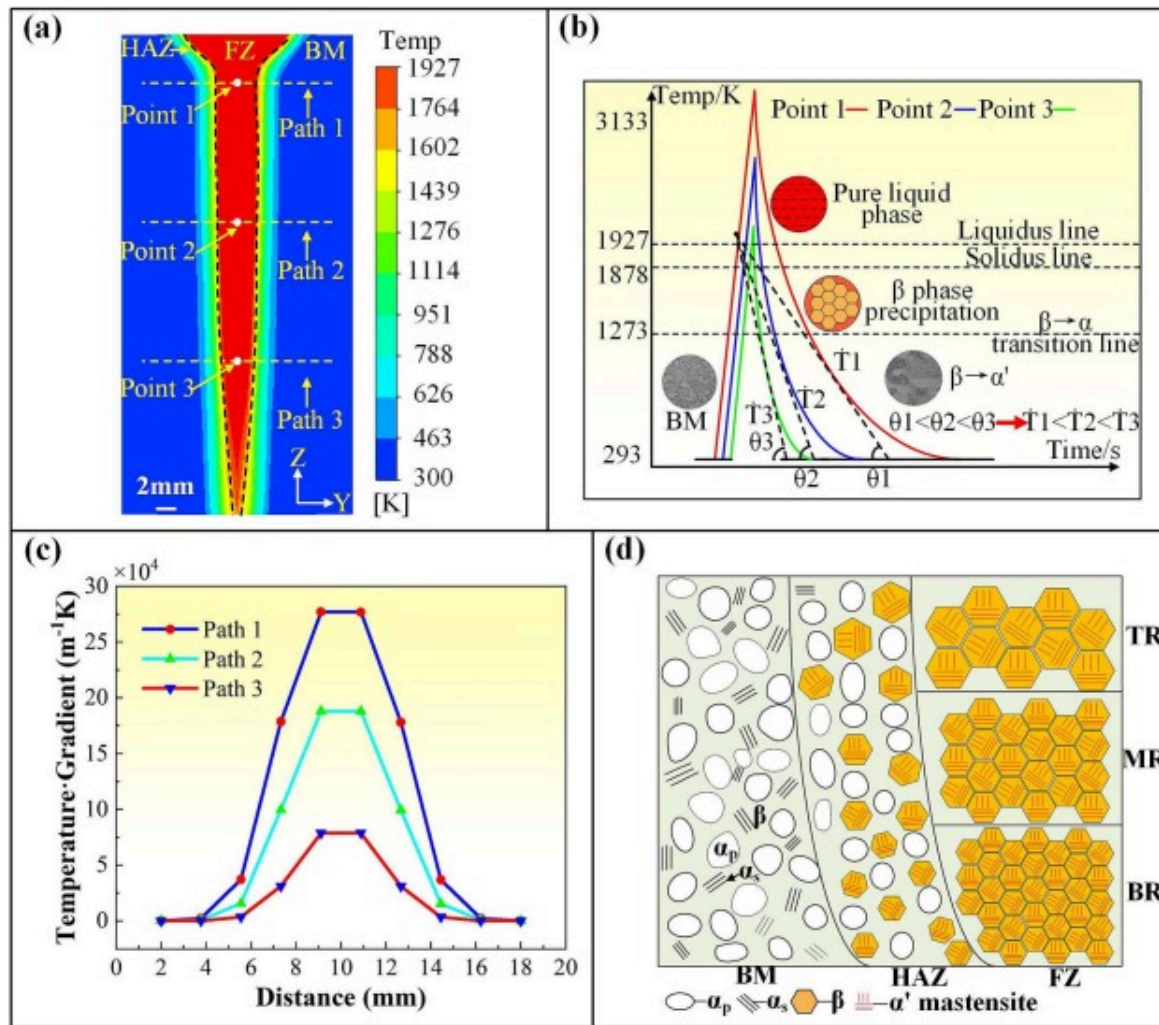
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Fig. 16. The flow conditions of the longitudinal section of the VLBW process molten pool at different times: (a) $t=50\text{ms}$; (b) $t=100\text{ms}$; (c) $t=200\text{ms}$; (d) $t=250\text{ms}$; (e) $t=300\text{ms}$; (f) $t=400\text{ms}$.

4.5. Mechanism analysis

During the VLBW process of TC4 titanium alloy thick plates, the heat input and cooling rates in different regions led to significant differences in microstructure and mechanical properties. To gain a deeper understanding of the influence of heat input in different regions on the microstructure of the weld region, this study systematically analyzed the effects of temperature gradient and cooling rate on the microstructure of different weld regions. And its microstructure evolution characteristics were further explained, as shown in [Fig. 17](#).



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Fig. 17. Microstructure evolution and mechanism analysis of different regions of weld seams: (a) The temperature distribution of the weld cross-section and the corresponding sampling positions in Fig. 17, Fig. 17; (b) Analysis of temperature evolution behavior and phase transition microstructure at different positions; (c) Temperature gradient curves

under different paths; (d) The mechanism diagrams of tissue evolution and distribution in different regions and locations.[21].

Fig. 17(a) shows the simulation results of the temperature field distribution of the weld cross-section. It can be seen that the weld area can be divided into three parts: the BM, the HAZ and the FZ. To analyze the temperature gradient and cooling rate differences in different areas, three representative positions, namely the TR, MR and BR of the weld seam, were selected on the same cross-section. Three paths (Path 1, Path 2 and Path 3) and the corresponding measurement points (Point 1, Point 2 and Point 3) were set respectively. The temperature gradient variation in the Y direction is extracted through the path data, and the time–temperature variation curves at each position are obtained through the measurement points. The relevant measurement results are shown in Fig. 17, Fig. 16.

Fig. 17(b) shows the time–temperature evolution curves at different positions and the corresponding phase transition processes. These three curves respectively represent the temperature changes over time at points 1, 2 and 3. It can be seen that each measurement point has undergone a complete thermal cycle process from solid-phase heating to liquid-phase cooling and solidification. It is worth noting that the peak temperatures at the three positions gradually decrease, which reflects the difference in the heat input distribution of TC4 thick plates during the VLBW process. The angle θ marked in the figure represents the inclination angles corresponding to the slopes of the temperature curves at different points at the same temperature. To obtain these slopes, tangent lines were drawn from a reference point, providing a straightforward and intuitive way to assess and compare the variation trends of the curves. This method is particularly useful in cases where direct derivative calculations are difficult or impractical, and it offers a reasonable approximation for capturing the general characteristics of curve variation. $\theta_3 > \theta_2 > \theta_1$ indicates the maximum cooling rate at point 3 (T) and the minimum cooling rate at point 1. Fig. 17(c) shows the variation of temperature gradients measured along Path 1, Path 2 and Path 3 with distance. The results show that the temperature gradient of the weld seam gradually decreases from TR to BR. For each path, the temperature gradient

gradually increases from BM to FZ. Especially in the heat-affected zone, there is a significant increase in the temperature gradient. This trend leads to the outward growth of grains from the edge of the fusion zone to the center of the molten pool.

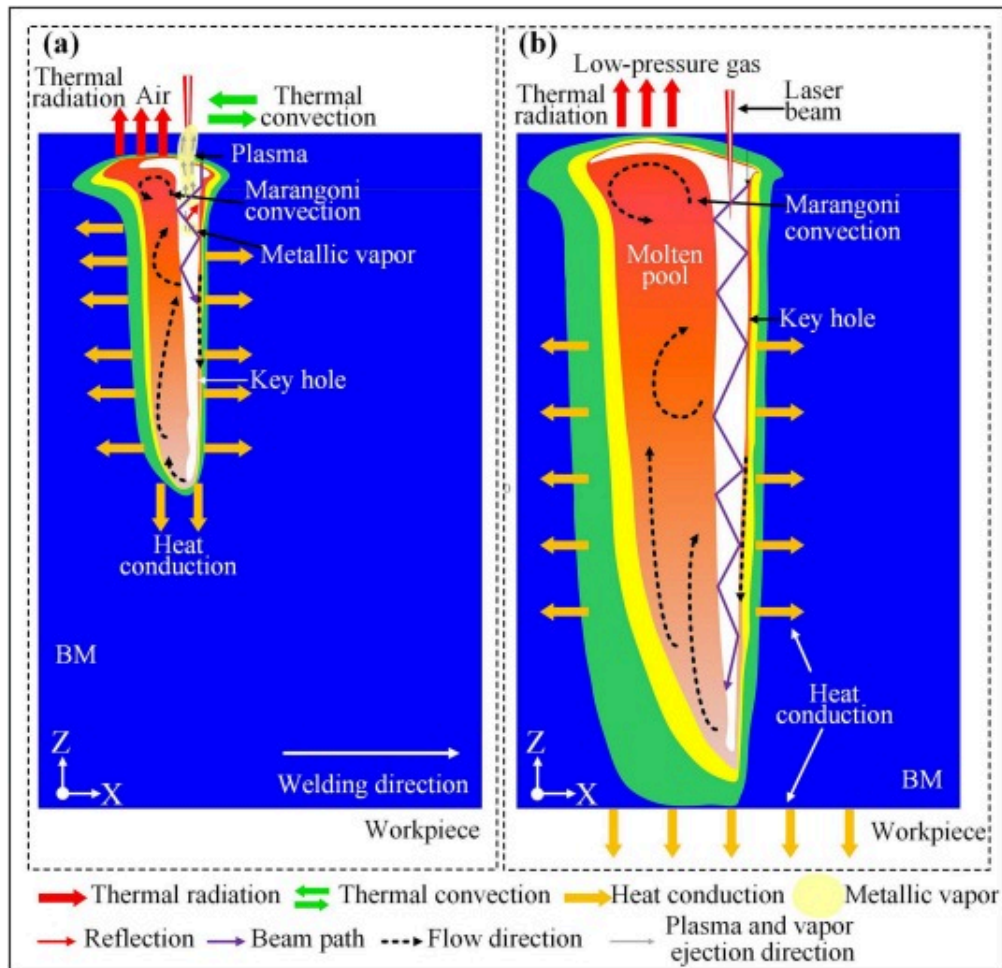
It is generally believed that the magnitude of the temperature gradient mainly determines the growth direction and morphology of the crystal, while the cooling rate dominates the coarseness of the grain size.[44,45] In a vacuum environment, since convective heat transfer is almost suppressed and heat is mainly dissipated through heat conduction, although the TR of the weld seam has a relatively high temperature gradient, the cooling rate is relatively low. The BR of the weld seam shows a faster cooling rate due to the higher difference in heat flux density. Therefore, in VLBW of TC4 thick plates, the grain evolution exhibits a clear spatial distribution: from top to bottom, the grains remain predominantly columnar, while their size gradually decreases. This refinement from TR to BR reflects the significant influence of variations in heat transfer and cooling rate under vacuum conditions on grain size. EBSD analysis further reveals that the α -phase in TR, MR, and BR is primarily aligned along the $\{0001\}$ and $\{11\bar{2}0\}$ planes, with the strongest texture observed in the TR. The texture strength gradually decreases from TR to BR, indicating that although the grains are refined toward the bottom, their overall morphology largely maintains the columnar growth direction. These observations confirm that the columnar grains preserve their preferential orientation along the heat flow direction, while the lower temperature gradient and faster cooling rate near the weld bottom, due to heat conduction to the substrate, primarily drive the reduction in grain size toward the BR.

Based on the above results, the microstructural evolution mechanisms in different regions of VLBW TC4 titanium alloy thick plates were analyzed, as illustrated in Fig. 17(d). The weld cross-section includes three distinct regions: the BM, the HAZ, and the FZ. In the BM region, which does not experience thermal cycling, the microstructure remains unchanged, primarily composed of primary α (α_p) and β phases. The HAZ, located between the BM and FZ, undergoes phase transformation and recrystallization due to thermal exposure. Near the BM side, only a small portion of the α_p phase transforms and

secondary α (α_s) precipitates during cooling. Closer to the FZ, with increased heat input, a larger fraction of α_p transforms into β during heating, followed by the formation of α_s and martensitic α' phases upon cooling. In the FZ, the material is fully melted by the high-energy laser beam, and all α phases transform into β during heating. Upon cooling, the β phase transforms into acicular α' phase, forming a typical basketweave structure. Additionally, due to the strong temperature gradient, coarse columnar grains grow preferentially along the thermal gradient direction.

According to the spatial distribution of heat input, the weld zone can be further divided into the TR, MR, and BR. Analysis of Fig. 17(b) indicates that the cooling rate increases gradually from TR to BR, leading to a corresponding decrease in grain size an observation consistent in both the HAZ and FZ. In the TR, where the heat input is concentrated and the cooling rate is the lowest, coarse grains are likely to form. In contrast, the BR experiences reduced heat input and enhanced heat conduction due to its proximity to the bottom surface, resulting in a significantly higher cooling rate and refined grain structure.

Fig. 18(a) and Fig. 18(b) illustrate the differences in energy transfer pathways and molten pool flow behaviors during laser welding of TC4 titanium alloy thick plates under atmospheric and vacuum environments, respectively. Under atmospheric condition (Fig. 18(a)), the incident laser beam interacts rapidly with the evaporated metallic vapor within the keyhole, leading to the formation of a high-density plasma and an upward-directed vapor plume. This plume partially obstructs the laser beam path, resulting in reflection and scattering of the laser energy, thereby reducing the overall energy coupling efficiency. Furthermore, laser-induced high-temperature gases give rise to strong thermal convection, exacerbating flow instability in both the keyhole and the molten pool regions, which hinders the consistent control of weld penetration. The molten pool also experiences vigorous Marangoni convection, driven by surface tension gradients and vapor recoil pressure, promoting chaotic surface flow and potentially affecting the uniformity of the weld morphology.



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Fig. 18. The influence of environmental pressure on energy transmission and molten pool behavior within the keyhole: (a) Atmospheric environment; (b) Vacuum environment.

In contrast, under vacuum conditions (Fig. 18(b)), the significantly reduced ambient pressure effectively suppresses the formation of a stable plasma plume, allowing the laser to penetrate more directly and efficiently along the full depth of the keyhole. This enhances energy transfer efficiency and promotes a more stable aspect ratio. At the same

time, the absence of natural convection in a vacuum reduces the cooling rate, leading to coarser grains compared to those formed under atmospheric conditions, which may to some extent weaken the mechanical properties of the weld. Although coarse grain formation is unfavorable for mechanical performance, under the same welding parameters, the vacuum environment markedly reduces plasma plume generation and limits the ingress of active elements such as N and O. This minimizes porosity and inclusions within the weld, resulting in a more stable and denser deep-penetration joint. Therefore, compared with the influence of grain coarsening, the effect of plasma plume suppression plays a more decisive role in improving the mechanical properties of the weld.

This comparative analysis highlights the fundamental differences in energy transfer mechanisms, keyhole stability, and molten pool dynamics between atmospheric and vacuum laser welding processes. It demonstrates that vacuum welding offers notable advantages in minimizing energy losses, stabilizing melt pool geometry, and tailoring weld microstructures—thereby providing a more favorable processing window for the high-quality joining of thick titanium alloy components.

5. Conclusion

This study systematically investigated the microstructural evolution, mechanical properties, and molten pool flow characteristics of TC4 titanium alloy thick plates during VLBW. By analyzing the microstructure, phase transformation behavior, and mechanical performance across the TR, MR, and BR of the weld, in combination with CFD numerical simulations, the mechanisms of keyhole evolution and molten pool dynamics under vacuum conditions were elucidated. The main conclusions are as follows:

- (1) The FZ consists of columnar grains with a basket-weave microstructure of acicular α' martensite and dense α'/α' interfaces, with the average grain size decreasing from 2.59 μm in the TR to 2.12 μm in the BR.

- (2) VLBW welds exhibit superior mechanical properties, as TR, MR, and BR specimens all fractured on the BM side, and maximum hardness reached 371 Hv, significantly higher than the BM (330 Hv).
- (3) Under vacuum conditions, the molten pool flow is primarily driven by recoil pressure and thermal buoyancy. This flow, coupled with heat conduction to the substrate, enhances temperature differences within the molten pool, resulting in varying cooling rates and temperature gradients.
- (4) VLBW slows heat dissipation but, with reduced plasma plume, enhances laser absorption, leading to deeper penetration, fewer defects, and superior weld quality in TC4 titanium alloy.

Data availability.

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CRediT authorship contribution statement

Jiwen Li: Writing – original draft. **Jianfeng Wang:** Writing – review & editing, Funding acquisition. **Zhan Liu:** Investigation. **Xingyue Lyu:** Formal analysis. **Xing Liu:** Methodology. **Zhiyong Li:** Software. **Xiaohong Zhan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Natural Science Foundation of China (Grant Nos. 52575398 and 52205380), the Natural Science Foundation of Jiangsu Province of China (Grant No. BK20220900), the China Postdoctoral Science Foundation (Grant No. 2022M721595), the Guangdong Basic and Applied Basic Research Foundation (Grant Nos. 2024A1515011083 and 2025A1515012762), the Shenzhen Science and Technology Program (Grant No. JCYJ20240813152503005), the Shenzhen Longhua Science and Technology Innovation Special Funding Project (Industrial Sci-Tech Innovation Center of Low-Altitude Intelligent Networking), the Young Elite Scientists Sponsorship Program by Jiangsu Association for Science and Technology (Grant No. JSTJ-2024-023), the Fundamental Research Funds for the Central Universities (Grant No. NS2024040), Jinan Science and Technology Program (Grant No. 202428046), and the Postgraduate Research & Practice Innovation Program of NUAA (Grant No. xcxjh20240606). J.F. Wang also acknowledged the support from the HORIZON EUROPE Marie Skłodowska-Curie Actions (Grant No. 101109931).

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

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